Week 4

1. Which Cassandra replication strategy is specifically designed for multi-datacenter deployments?

A) SimpleStrategy
B) NetworkTopologyStrategy
C) RandomPartitioner
D) ByteOrderedPartitioner

Answer: B

Explanation

A. SimpleStrategy (Incorrect): `SimpleStrategy` is intended for single-datacenter environments and is not recommended for production multi-datacenter setups.

B. NetworkTopologyStrategy (Correct): `NetworkTopologyStrategy` is designed for multi-datacenter deployments and allows different replication factors to be configured for each datacenter.

C. RandomPartitioner (Incorrect): RandomPartitioner determines how data is distributed across nodes and is unrelated to replication strategy.

D. ByteOrderedPartitioner (Incorrect): ByteOrderedPartitioner controls data ordering based on key values and does not manage replication across datacenters.

2. What is the primary purpose of Cassandra's Bloom Filter?

A) Store key-value pairs permanently
B) Compress SSTables
C) Quickly determine if a key may exist in an SSTable
D) Replace indexing structures

Answer: C

Explanation

A. Store key-value pairs permanently (Incorrect): Data is permanently stored in MemTables and SSTables, not in Bloom Filters.

B. Compress SSTables (Incorrect): SSTable compression is handled by compression mechanisms, whereas Bloom Filters are used for fast key existence checks.

C. Quickly determine if a key may exist in an SSTable (Correct): A Bloom Filter helps Cassandra avoid unnecessary disk reads by checking whether a key might be present in an SSTable before searching it.

D. Replace indexing structures (Incorrect): Bloom Filters complement indexes by reducing unnecessary lookups; they do not replace indexing structures.

3. Which statement about Bloom Filters is TRUE?

A) They may produce false negatives.
B) They never produce false positives.
C) They may produce false positives but never false negatives.
D) They guarantee exact membership testing.

Answer: C

Explanation

A. They may produce false negatives (Incorrect): Bloom Filters are designed to avoid false negatives; an existing element will never be reported as absent.

B. They never produce false positives (Incorrect): Due to hash collisions, Bloom Filters can occasionally report that a non-existing element is present.

C. They may produce false positives but never false negatives (Correct): A Bloom Filter may incorrectly indicate that an element exists, but if it says an element is not present, that result is always correct.

D. They guarantee exact membership testing (Incorrect): Bloom Filters provide probabilistic membership testing and do not guarantee exact results because false positives are possible.

4. In Cassandra, why are writes generally faster than reads?

A) Reads use only memory.
B) Writes avoid disk operations entirely.
C) Writes are append-oriented using Commit Logs and MemTables.
D) Reads bypass SSTables.

Answer: C

Explanation

A. Reads use only memory (Incorrect): Reads may require checking MemTables, caches, and multiple SSTables on disk, not just memory.

B. Writes avoid disk operations entirely (Incorrect): Every write is recorded in the Commit Log on disk for durability before acknowledgment.

C. Writes are append-oriented using Commit Logs and MemTables (Correct): Cassandra first writes data sequentially to the Commit Log and then to an in-memory MemTable, making write operations very efficient.

D. Reads bypass SSTables (Incorrect): Reads often need to access SSTables, especially when the requested data is not available in memory.

5. Which Cassandra consistency level is the FASTEST for writes?

A) ALL
B) QUORUM
C) ONE
D) ANY

Answer: D

Explanation

A. ALL (Incorrect): `ALL` requires acknowledgment from all replicas, making it the slowest but most consistent write option.

B. QUORUM (Incorrect): `QUORUM` requires acknowledgment from a majority of replicas, providing a balance between consistency and performance.

C. ONE (Incorrect): `ONE` requires acknowledgment from at least one replica and is fast, but `ANY` can be even faster since a hint can satisfy the write request when replicas are unavailable.

D. ANY (Correct): `ANY` is the fastest write consistency level because Cassandra can acknowledge the write even if no replica is currently available, by storing a hint for later delivery.

6. According to CAP theorem, Cassandra primarily chooses:

A) Consistency and Availability
B) Consistency and Partition Tolerance
C) Availability and Partition Tolerance
D) Atomicity and Durability

Answer: C

Explanation

A. Consistency and Availability (Incorrect): CAP theorem states that during a network partition, a distributed system cannot guarantee both consistency and availability simultaneously.

B. Consistency and Partition Tolerance (Incorrect): Systems that prioritize consistency during partitions may sacrifice availability; Cassandra generally favors availability.

C. Availability and Partition Tolerance (Correct): Cassandra is designed to remain available even during network partitions, making it an AP (Availability + Partition Tolerance) system in the CAP theorem.

D. Atomicity and Durability (Incorrect): Atomicity and Durability are ACID properties and are not the components of the CAP theorem.

7. Which property defines Eventual Consistency?

A) All replicas are instantly synchronized.
B) Replicas converge if updates stop.
C) Reads always return latest value.
D) No stale reads are possible.

Answer: B

Explanation

A. All replicas are instantly synchronized (Incorrect): Eventual consistency allows temporary differences between replicas; synchronization is not immediate.

B. Replicas converge if updates stop (Correct): Eventual consistency guarantees that if no new updates occur, all replicas will eventually become consistent and contain the same data.

C. Reads always return the latest value (Incorrect): A read may return stale data until all replicas have converged.

D. No stale reads are possible (Incorrect): Stale reads can occur in eventually consistent systems before replica synchronization is complete.

8. What happens if a ZooKeeper ensemble loses majority availability?

A) A new leader is elected immediately.
B) Remaining node becomes leader.
C) No leader can be elected.
D) System switches to standalone mode.

Answer: C)

Explanation

A. A new leader is elected immediately (Incorrect): A leader can only be elected if a majority of ZooKeeper nodes are available.

B. Remaining node becomes leader (Incorrect): A single remaining node cannot become leader without quorum approval from the majority of the ensemble.

C. No leader can be elected (Correct): ZooKeeper requires a majority (quorum) of nodes to elect and maintain a leader. Without a quorum, leader election cannot occur and the ensemble cannot process write requests.

D. System switches to standalone mode (Incorrect): ZooKeeper does not automatically switch a distributed ensemble into standalone mode when quorum is lost.

9. Which ZooKeeper design goal ensures that all updates are processed in a globally agreed order?

A) Ordered
B) Replicated
C) Fast
D) Atomic

Answer: A

Explanation

A. Ordered (Correct): ZooKeeper guarantees that all updates are processed in a consistent global order across the ensemble, ensuring all clients observe changes in the same sequence.

B. Replicated (Incorrect): Replication ensures data is copied across multiple servers for fault tolerance, but it does not by itself guarantee global ordering of updates.

C. Fast (Incorrect): Fast refers to ZooKeeper's ability to provide low-latency coordination services, not the ordering of updates.

D. Atomic (Incorrect): Atomicity ensures an update either succeeds completely or fails completely, but it does not define the global order in which updates are processed.

10. What is the default type of znode in ZooKeeper?

A) Ephemeral
B) Sequential
C) Temporary
D) Persistent

Answer: D

Explanation

A. Ephemeral (Incorrect): Ephemeral znodes exist only as long as the creating client's session is active and are not the default type.

B. Sequential (Incorrect): Sequential is an optional attribute that appends a unique sequence number to a znode's name; it is not the default type.

C. Temporary (Incorrect): ZooKeeper does not have a znode type called "Temporary"; the correct temporary-like type is Ephemeral.

D. Persistent (Correct): By default, a znode is persistent, meaning it remains in ZooKeeper until it is explicitly deleted, even if the client that created it disconnects**.**